

NETWORK IMPLEMENTATION & VERIFICATION REPORT

Team 5

Giovanni Emil 2024/05404

Mina Samaan Zakaria 2024/04830

Youssef Mohamed Ahmed Ibrahim Abdelaal 2024/05244

Selim Mamdouh 2024/03541

Mohab Ibrahim 2024/03899

Instructor:

Dr. Yasmin AbdAlhamid

VLSM TABLE:

Subnet Description	VLAN ID	Hosts	Network/CIDR	First Usable	Last Usable	Broadcast
VLAN 80	80	61	10.0.0.0/26	10.0.0.1	10.0.0.62	10.0.0.63
VLAN 40	40	45	10.0.0.64/26	10.0.0.65	10.0.0.126	10.0.0.127
VLAN 130	130	31	10.0.0.128/26	10.0.0.129	10.0.0.190	10.0.0.191
VLAN 60	60	30	10.0.0.192/27	10.0.0.193	10.0.0.222	10.0.0.223
VLAN 70	70	25	10.0.0.224/27	10.0.0.225	10.0.0.254	10.0.0.255
VLAN 50	50	15	10.0.1.0/27	10.0.1.1	10.0.1.30	10.0.1.31
VLAN 10	10	12	10.0.1.32/28	10.0.1.33	10.0.1.46	10.0.1.47
VLAN 20	20	7	10.0.1.48/28	10.0.1.49	10.0.1.62	10.0.1.63
From SW-S to MIU-MIU-GW	—	7	10.0.1.64/28	10.0.1.65	10.0.1.78	10.0.1.79
From Main-MLS to MIU-MIU-GW	—	2	10.0.1.80/30	10.0.1.81	10.0.1.82	10.0.1.83
From N-MLS to MIU-MIU-GW	—	2	10.0.1.84/30	10.0.1.85	10.0.1.86	10.0.1.87
From S-MLS to MIU-MIU-GW	—	2	10.0.1.88/30	10.0.1.89	10.0.1.90	10.0.1.91
From R-MLS to MIU-MIU-GW	—	2	10.0.1.92/30	10.0.1.93	10.0.1.94	10.0.1.95

ADDRESSING TABLE:

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	VLAN ID
Main-MLS	G1/1/1	10.0.1.81	255.255.255.252	—	—
Main-MLS	VLAN 10	10.0.1.33	255.255.255.240	—	10
Main-MLS	VLAN 20	10.0.1.49	255.255.255.240	—	20
S-MLS	G1/1/1	10.0.1.89	255.255.255.252	—	—
S-MLS	VLAN 130	10.0.0.129	255.255.255.192	—	130
S-MLS	VLAN 40	10.0.0.65	255.255.255.192	—	40
N-MLS	G1/1/1	10.0.1.85	255.255.255.252	—	—
N-MLS	VLAN 50	10.0.1.1	255.255.255.224	—	50
N-MLS	VLAN 60	10.0.0.193	255.255.255.224	—	60
R-MLS	G1/0/5	10.0.1.93	255.255.255.252	—	—
R-MLS	VLAN 70	10.0.0.225	255.255.255.224	—	70
R-MLS	VLAN 80	10.0.0.1	255.255.255.192	—	80
MIU-GW	Gig0/0	209.165.200.226	255.255.255.240	—	—
MIU-GW	Gig0/1	10.0.1.94	255.255.255.252	—	—
MIU-GW	Gig0/0/0	10.0.1.66	255.255.255.240	—	—
MIU-GW	Gig0/1/0	10.0.1.82	255.255.255.252	—	—
MIU-GW	Gig0/2/0	10.0.1.90	255.255.255.252	—	—
MIU-GW	Gig0/3/0	10.0.1.86	255.255.255.252	—	—
ISP	Gig0/0	209.165.200.225	255.255.255.240	—	—
ISP	Gig0/1	64.100.1.1	255.255.255.224	—	—
ISP	Gig0/2	64.100.2.1	255.255.255.224	—	—
Branch-GW	Gig0/0.2	192.168.2.1	255.255.255.0	—	2
Branch-GW	Gig0/0.3	192.168.3.1	255.255.255.0	—	3
Branch-GW	Gig0/1	64.100.1.2	255.255.255.224	—	—
Router	G0/1	64.100.2.2	255.255.255.224	64.100.2.1	—
Router	Router IP	192.168.10.1	255.255.255.128	—	—

HOSTS ADDRESS TABLE:

PC / Device	VLAN ID	IP Address	Subnet Mask	Default Gateway
PC-1	10	From DHCP Server	From DHCP Server	From DHCP Server
PC-2	10	From DHCP Server	From DHCP Server	From DHCP Server
PC-3	20	From DHCP Server	From DHCP Server	From DHCP Server
PC-4	130	From DHCP Server	From DHCP Server	From DHCP Server
PC-5	130	From DHCP Server	From DHCP Server	From DHCP Server
PC-6	40	From DHCP Server	From DHCP Server	From DHCP Server
PC-7	50	10.0.1.2	255.255.255.224	10.0.1.1
PC-8	60	10.0.0.194	255.255.255.224	10.0.0.193
PC-9	70	10.0.0.226	255.255.255.224	10.0.0.225
PC-10	80	10.0.0.2	255.255.255.192	10.0.0.1
PC-11	2	From DHCP Server	From DHCP Server	From DHCP Server
PC-12	3	From DHCP Server	From DHCP Server	From DHCP Server
Laptop	—	192.168.10.10	255.255.255.128	192.168.10.1
Tablet	—	192.168.10.30	255.255.255.128	192.168.10.1
Smartphone	—	192.168.10.20	255.255.255.128	192.168.10.1
DHCP Server	—	10.0.1.67	255.255.255.240	10.0.1.66
Email Server	—	10.0.1.68	255.255.255.240	10.0.1.66
Web Server	—	10.0.1.69	255.255.255.240	10.0.1.66
DNS Server	—	10.0.1.70	255.255.255.240	10.0.1.66
NTP & Syslog Server	—	10.0.1.71	255.255.255.240	10.0.1.66

CONFUGRATIONS:

Main-MLS: Building configuration...

Current configuration : 2725 bytes

!

version 16.3.2

service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Main-MLS

!

!

no profinet

!

```
!  
!  
!  
!  
!  
no ip cef  
ip routing  
!  
no ipv6 cef  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
no ip domain-lookup  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface Port-channel2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
!  
interface Port-channel3  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
!  
interface GigabitEthernet1/0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
channel-group 2 mode active
```

```
!  
interface GigabitEthernet1/0/2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
channel-group 2 mode active  
!  
interface GigabitEthernet1/0/3  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
channel-group 3 mode active  
!  
interface GigabitEthernet1/0/4  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
channel-group 3 mode active  
!  
interface GigabitEthernet1/0/5  
!  
interface GigabitEthernet1/0/6  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16  
!  
interface GigabitEthernet1/0/17  
!  
interface GigabitEthernet1/0/18
```

```
!  
interface GigabitEthernet1/0/19  
!  
interface GigabitEthernet1/0/20  
!  
interface GigabitEthernet1/0/21  
!  
interface GigabitEthernet1/0/22  
!  
interface GigabitEthernet1/0/23  
!  
interface GigabitEthernet1/0/24  
!  
interface GigabitEthernet1/1/1  
no switchport  
ip address 10.0.1.81 255.255.255.252  
!  
interface GigabitEthernet1/1/2  
!  
interface GigabitEthernet1/1/3  
!  
interface GigabitEthernet1/1/4  
!  
interface Vlan1  
no ip address  
shutdown  
!  
interface Vlan10  
mac-address 00e0.b08c.5b01  
ip address 10.0.1.33 255.255.255.240  
ip helper-address 10.0.1.82  
!  
interface Vlan20  
mac-address 00e0.b08c.5b02  
ip address 10.0.1.49 255.255.255.240  
ip helper-address 10.0.1.82  
!  
router ospf 100  
router-id 2.2.2.2  
log-adjacency-changes  
passive-interface Vlan10  
passive-interface Vlan20  
network 10.0.1.80 0.0.0.3 area 0  
network 10.0.1.32 0.0.0.15 area 0  
network 10.0.1.48 0.0.0.15 area 0  
!
```

```
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^C Authorized Access Only ^C
!
!
!
logging trap debugging
logging 10.0.1.71
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
ntp server 10.0.1.71
!
End
```

Main-SW1: Building configuration...

```
Current configuration : 1954 bytes
!
version 15.0
service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Main-SW1
!
!
!
!
!
!
spanning-tree mode pvst
```



```
spanning-tree extend system-id
!
interface Port-channel1
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20
switchport mode trunk
!
interface Port-channel2
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/3
switchport access vlan 10
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
```

```
!  
interface FastEthernet0/13  
!  
interface FastEthernet0/14  
!  
interface FastEthernet0/15  
!  
interface FastEthernet0/16  
!  
interface FastEthernet0/17  
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24  
!  
interface GigabitEthernet0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
channel-group 2 mode active  
!  
interface GigabitEthernet0/2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
channel-group 2 mode active  
!  
interface Vlan1  
no ip address  
shutdown  
!  
logging trap debugging  
logging 10.0.1.71  
!  
!  
!
```

```
line con 0
!  
line vty 0 4  
login  
line vty 5 15  
login  
!  
!  
!  
ntp server 10.0.1.71  
!  
End
```

Main-SW2 : Building configuration...

```
Current configuration : 2029 bytes  
!  
version 15.0  
service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Main-SW2  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface Port-channel1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
!  
interface Port-channel3  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 10,20  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 1000
```

```
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/3
switchport access vlan 10
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/4
switchport access vlan 20
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
```

```
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 3 mode active
!
interface GigabitEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 10,20
switchport mode trunk
channel-group 3 mode active
!
interface Vlan1
no ip address
shutdown
!
logging trap debugging
logging 10.0.1.71
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
ntp server 10.0.1.71
!
```

End

S-MLS : Building configuration...

Current configuration : 2770 bytes

!

version 16.3.2

service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname S-MLS

!

!

no profinet

!

!

!

!

!

!

!

no ip cef

ip routing

!

no ipv6 cef

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

spanning-tree mode pvst

!

!

!

!

!

!

!

```
interface Port-channel2
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
!
interface GigabitEthernet1/0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
channel-group 2 mode active
!
interface GigabitEthernet1/0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
channel-group 2 mode active
!
interface GigabitEthernet1/0/3
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
channel-group 3 mode active
!
interface GigabitEthernet1/0/4
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
channel-group 3 mode active
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
```

```
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
no switchport
ip address 10.0.1.89 255.255.255.252
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
interface Vlan40
mac-address 0030.f2dd.7801
ip address 10.0.0.65 255.255.255.192
ip helper-address 10.0.1.67
```



```
ip helper-address 10.0.1.90
!
interface Vlan130
mac-address 0030.f2dd.7802
ip address 10.0.0.129 255.255.255.192
ip helper-address 10.0.1.67
ip helper-address 10.0.1.90
!
router ospf 100
router-id 3.3.3.3
log-adjacency-changes
passive-interface Vlan40
passive-interface Vlan130
network 10.0.1.88 0.0.0.3 area 0
network 10.0.1.64 0.0.0.63 area 0
network 10.0.0.128 0.0.0.63 area 0
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^C Authorized Access Only ^C
!
!
!
!
logging trap debugging
logging 10.0.1.71
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
ntp server 10.0.1.71
!
End
```

S-SW1:

Current configuration : 1958 bytes

!

```
version 15.0
service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S-SW1
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel1
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
!
interface Port-channel2
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/3
switchport access vlan 130
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
```

```
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 40,130
switchport mode trunk
channel-group 2 mode active
!
interface GigabitEthernet0/2
switchport trunk native vlan 1000
```

```
switchport trunk allowed vlan 40,130
switchport mode trunk
channel-group 2 mode active
!
interface Vlan1
no ip address
shutdown
!
logging trap debugging
logging 10.0.1.71
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
ntp server 10.0.1.71
!
end
```

S-SW2: Building configuration...

```
Current configuration : 2033 bytes
!
version 15.0
service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S-SW2
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
```

```
!  
interface Port-channel1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 40,130  
switchport mode trunk  
!  
interface Port-channel3  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 40,130  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 40,130  
switchport mode trunk  
channel-group 1 mode active  
!  
interface FastEthernet0/2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 40,130  
switchport mode trunk  
channel-group 1 mode active  
!  
interface FastEthernet0/3  
switchport access vlan 130  
switchport mode access  
spanning-tree portfast  
!  
interface FastEthernet0/4  
switchport access vlan 40  
switchport mode access  
spanning-tree portfast  
!  
interface FastEthernet0/5  
!  
interface FastEthernet0/6  
!  
interface FastEthernet0/7  
!  
interface FastEthernet0/8  
!  
interface FastEthernet0/9  
!  
interface FastEthernet0/10  
!  
interface FastEthernet0/11
```

```
!  
interface FastEthernet0/12  
!  
interface FastEthernet0/13  
!  
interface FastEthernet0/14  
!  
interface FastEthernet0/15  
!  
interface FastEthernet0/16  
!  
interface FastEthernet0/17  
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24  
!  
interface GigabitEthernet0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 40,130  
switchport mode trunk  
channel-group 3 mode active  
!  
interface GigabitEthernet0/2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 40,130  
switchport mode trunk  
channel-group 3 mode active  
!  
interface Vlan1  
no ip address  
shutdown  
!  
logging trap debugging  
logging 10.0.1.71  
!
```

```
!  
!  
line con 0  
!  
line vty 0 4  
login  
line vty 5 15  
login  
!  
!  
!  
ntp server 10.0.1.71  
!  
End
```

N-MLS : Building configuration...

```
Current configuration : 2581 bytes  
!  
version 16.3.2  
service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname N-MLS  
!  
!  
no profinet  
!  
!  
!  
!  
!  
!  
no ip cef  
ip routing  
!  
no ipv6 cef  
!  
!  
!  
!  
!  
!  
!
```

```
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface Port-channel2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
!  
interface Port-channel3  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
!  
interface GigabitEthernet1/0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
channel-group 2 mode active  
!  
interface GigabitEthernet1/0/2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
channel-group 2 mode active  
!  
interface GigabitEthernet1/0/3  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
channel-group 3 mode active  
!  
interface GigabitEthernet1/0/4  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
channel-group 3 mode active
```



```
!  
interface GigabitEthernet1/0/5  
!  
interface GigabitEthernet1/0/6  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16  
!  
interface GigabitEthernet1/0/17  
!  
interface GigabitEthernet1/0/18  
!  
interface GigabitEthernet1/0/19  
!  
interface GigabitEthernet1/0/20  
!  
interface GigabitEthernet1/0/21  
!  
interface GigabitEthernet1/0/22  
!  
interface GigabitEthernet1/0/23  
!  
interface GigabitEthernet1/0/24  
!  
interface GigabitEthernet1/1/1  
no switchport  
ip address 10.0.1.85 255.255.255.252  
!  
interface GigabitEthernet1/1/2
```

```
!  
interface GigabitEthernet1/1/3  
!  
interface GigabitEthernet1/1/4  
!  
interface Vlan1  
no ip address  
shutdown  
!  
interface Vlan50  
mac-address 0060.2f40.9601  
ip address 10.0.1.1 255.255.255.224  
!  
interface Vlan60  
mac-address 0060.2f40.9602  
ip address 10.0.0.193 255.255.255.224  
!  
router eigrp 10  
passive-interface Vlan50  
passive-interface Vlan60  
network 10.0.1.84 0.0.0.3  
network 10.0.1.0 0.0.0.31  
network 10.0.0.192 0.0.0.31  
!  
ip classless  
!  
ip flow-export version 9  
!  
!  
!  
banner motd ^C Authorized Access Only ^C  
!  
!  
!  
!  
logging trap debugging  
logging 10.0.1.71  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login  
!  
!  
!
```

```
ntp server 10.0.1.71
!  
End
```

N-SW1: Building configuration...

```
Current configuration : 1951 bytes
!  
version 15.0  
service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname N-SW1  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface Port-channel1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
!  
interface Port-channel2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
channel-group 1 mode active  
!  
interface FastEthernet0/2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
channel-group 1 mode active
```

```
!  
interface FastEthernet0/3  
switchport access vlan 50  
switchport mode access  
spanning-tree portfast  
!  
interface FastEthernet0/4  
!  
interface FastEthernet0/5  
!  
interface FastEthernet0/6  
!  
interface FastEthernet0/7  
!  
interface FastEthernet0/8  
!  
interface FastEthernet0/9  
!  
interface FastEthernet0/10  
!  
interface FastEthernet0/11  
!  
interface FastEthernet0/12  
!  
interface FastEthernet0/13  
!  
interface FastEthernet0/14  
!  
interface FastEthernet0/15  
!  
interface FastEthernet0/16  
!  
interface FastEthernet0/17  
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!
```

```
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 50,60
switchport mode trunk
channel-group 2 mode active
!
interface GigabitEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 50,60
switchport mode trunk
channel-group 2 mode active
!
interface Vlan1
no ip address
shutdown
!
logging trap debugging
logging 10.0.1.71
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
ntp server 10.0.1.71
!
End
```

N-SW2: Building configuration...

```
Current configuration : 1951 bytes
!
version 15.0
service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname N-SW2
```

```
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface Port-channel1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
!  
interface Port-channel3  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
channel-group 1 mode active  
!  
interface FastEthernet0/2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 50,60  
switchport mode trunk  
channel-group 1 mode active  
!  
interface FastEthernet0/3  
switchport access vlan 60  
switchport mode access  
spanning-tree portfast  
!  
interface FastEthernet0/4  
!  
interface FastEthernet0/5  
!  
interface FastEthernet0/6  
!  
interface FastEthernet0/7  
!  
interface FastEthernet0/8  
!
```

```
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 50,60
switchport mode trunk
channel-group 3 mode active
!
interface GigabitEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 50,60
switchport mode trunk
channel-group 3 mode active
!
interface Vlan1
no ip address
```

```
shutdown
!
logging trap debugging
logging 10.0.1.71
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
ntp server 10.0.1.71
!
End
```

R-MLS: Building configuration...

```
Current configuration : 2601 bytes
!
version 16.3.2
service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-MLS
!
!
no profinet
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
!
!
!
```



```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface Port-channel2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
!  
interface Port-channel3  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
!  
interface GigabitEthernet1/0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
channel-group 2 mode active  
!  
interface GigabitEthernet1/0/2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
channel-group 2 mode active  
!  
interface GigabitEthernet1/0/3  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
channel-group 3 mode active  
!
```

```
interface GigabitEthernet1/0/4
switchport trunk native vlan 1000
switchport trunk allowed vlan 70,80
switchport mode trunk
channel-group 3 mode active
!
interface GigabitEthernet1/0/5
no switchport
ip address 10.0.1.93 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
```

```
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
interface Vlan70
mac-address 0010.1112.c501
ip address 10.0.0.225 255.255.255.224
!
interface Vlan80
mac-address 0010.1112.c502
ip address 10.0.0.1 255.255.255.192
!
router eigrp 10
passive-interface Vlan70
passive-interface Vlan80
network 10.0.1.92 0.0.0.3
network 10.0.0.224 0.0.0.31
network 10.0.0.0 0.0.0.63
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^C Authorized Access ^C
!
!
!
!
logging trap debugging
logging 10.0.1.71
line con 0
!
```

```
line aux 0
!  
line vty 0 4  
login  
!  
!  
!  
ntp server 10.0.1.71  
!  
End
```

R-SW1: Building configuration...

```
Current configuration : 1951 bytes  
!  
version 15.0  
service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname R-SW1  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface Port-channel1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
!  
interface Port-channel2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
channel-group 1 mode active  
!
```

```
interface FastEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 70,80
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/3
switchport access vlan 70
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
```

```
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24  
!  
interface GigabitEthernet0/1  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
channel-group 2 mode active  
!  
interface GigabitEthernet0/2  
switchport trunk native vlan 1000  
switchport trunk allowed vlan 70,80  
switchport mode trunk  
channel-group 2 mode active  
!  
interface Vlan1  
no ip address  
shutdown  
!  
logging trap debugging  
logging 10.0.1.71  
!  
!  
!  
line con 0  
!  
line vty 0 4  
login  
line vty 5 15  
login  
!  
!  
!  
ntp server 10.0.1.71  
!  
End
```

R-SW2: Building configuration...

Current configuration : 1927 bytes

```
!  
version 15.0
```

```
service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R-SW2
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface Port-channel1
switchport trunk native vlan 1000
switchport trunk allowed vlan 70,80
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 1000
switchport trunk allowed vlan 70,80
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 70,80
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 70,80
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/3
switchport access vlan 80
switchport mode access
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
```

```
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 70,80
switchport mode trunk
channel-group 3 mode active
!
interface GigabitEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 70,80
switchport mode trunk
```



```
channel-group 3 mode active
!  
interface Vlan1  
no ip address  
shutdown  
!  
logging trap debugging  
logging 10.0.1.71  
!  
!  
!  
line con 0  
!  
line vty 0 4  
login  
line vty 5 15  
login  
!  
!  
!  
ntp server 10.0.1.71  
!  
end
```

B_S1: Building configuration...

```
Current configuration : 1700 bytes  
!  
version 15.0  
service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname B-S1  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface Port-channel1  
switchport trunk native vlan 1000
```

```
switchport trunk allowed vlan 2-3
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 2-3
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/2
switchport trunk native vlan 1000
switchport trunk allowed vlan 2-3
switchport mode trunk
channel-group 1 mode active
!
interface FastEthernet0/3
switchport access vlan 2
switchport mode access
spanning-tree portfast
!
interface FastEthernet0/4
!
interface FastEthernet0/5
!
interface FastEthernet0/6
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
!
```

```
interface FastEthernet0/17
!
interface FastEthernet0/18
!
interface FastEthernet0/19
!
interface FastEthernet0/20
!
interface FastEthernet0/21
!
interface FastEthernet0/22
!
interface FastEthernet0/23
!
interface FastEthernet0/24
!
interface GigabitEthernet0/1
switchport trunk native vlan 1000
switchport trunk allowed vlan 2-3
switchport mode trunk
!
interface GigabitEthernet0/2
!
interface Vlan1
no ip address
shutdown
!
interface Vlan2
no ip address
!
logging trap debugging
logging 10.0.1.71
!
!
!
line con 0
!
line vty 0 4
login
line vty 5 15
login
!
!
!
ntp server 10.0.1.71
!
```

End

B_S2: Building configuration...

Current configuration : 1574 bytes

!

version 15.0

service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname B-S2

!

!

!

!

!

!

spanning-tree mode pvst

spanning-tree extend system-id

!

interface Port-channel1

switchport trunk native vlan 1000

switchport trunk allowed vlan 2-3

switchport mode trunk

!

interface FastEthernet0/1

switchport trunk native vlan 1000

switchport trunk allowed vlan 2-3

switchport mode trunk

channel-group 1 mode active

!

interface FastEthernet0/2

switchport trunk native vlan 1000

switchport trunk allowed vlan 2-3

switchport mode trunk

channel-group 1 mode active

!

interface FastEthernet0/3

switchport access vlan 3

switchport mode access

spanning-tree portfast

!

interface FastEthernet0/4

!

interface FastEthernet0/5

```
!  
interface FastEthernet0/6  
!  
interface FastEthernet0/7  
!  
interface FastEthernet0/8  
!  
interface FastEthernet0/9  
!  
interface FastEthernet0/10  
!  
interface FastEthernet0/11  
!  
interface FastEthernet0/12  
!  
interface FastEthernet0/13  
!  
interface FastEthernet0/14  
!  
interface FastEthernet0/15  
!  
interface FastEthernet0/16  
!  
interface FastEthernet0/17  
!  
interface FastEthernet0/18  
!  
interface FastEthernet0/19  
!  
interface FastEthernet0/20  
!  
interface FastEthernet0/21  
!  
interface FastEthernet0/22  
!  
interface FastEthernet0/23  
!  
interface FastEthernet0/24  
!  
interface GigabitEthernet0/1  
!  
interface GigabitEthernet0/2  
!  
interface Vlan1  
no ip address  
shutdown
```

```
!  
logging trap debugging  
logging 10.0.1.71  
!  
!  
!  
line con 0  
!  
line vty 0 4  
login  
line vty 5 15  
login  
!  
!  
!  
ntp server 10.0.1.71  
!  
End
```

Branch_GW: Building configuration...

Current configuration : 1897 bytes

```
!  
version 15.1  
service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
security passwords min-length 10  
!  
hostname Branch-GW  
!  
!  
!  
!  
ip dhcp excluded-address 192.168.2.1 192.168.2.5  
ip dhcp excluded-address 192.168.3.1 192.168.3.5  
!  
ip dhcp pool B-LAN2  
network 192.168.2.0 255.255.255.0  
default-router 192.168.2.1  
dns-server 10.0.1.70  
ip dhcp pool B-LAN3  
network 192.168.3.0 255.255.255.0  
default-router 192.168.3.1  
dns-server 10.0.1.70  
!
```

```
!  
!  
ip cef  
no ipv6 cef  
!  
!  
!  
!  
license udi pid CISCO2911/K9 sn FTX152408IY-  
license boot module c2900 technology-package securityk9  
!  
!  
!  
crypto isakmp policy 10  
encr aes 256  
authentication pre-share  
group 5  
!  
crypto isakmp key cisco123 address 209.165.200.226  
!  
!  
!  
crypto ipsec transform-set BRANCH_TS esp-aes 256 esp-sha-hmac  
!  
crypto map BRANCH_VPN_MAP 10 ipsec-isakmp  
set peer 209.165.200.226  
set transform-set BRANCH_TS  
match address VPN_ACL  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
no ip address  
duplex auto
```

```
speed auto
!
interface GigabitEthernet0/0.2
encapsulation dot1Q 2
ip address 192.168.2.1 255.255.255.0
!
interface GigabitEthernet0/0.3
encapsulation dot1Q 3
ip address 192.168.3.1 255.255.255.0
!
interface GigabitEthernet0/1
ip address 64.100.1.2 255.255.255.224
duplex auto
speed auto
crypto map BRANCH_VPN_MAP
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 0.0.0.0 0.0.0.0 64.100.1.1
!
ip flow-export version 9
!
!
ip access-list extended VPN_ACL
permit ip 192.168.2.0 0.0.0.255 10.0.0.0 0.0.1.255
permit ip 192.168.3.0 0.0.0.255 10.0.0.0 0.0.1.255
!
banner motd ^C Authorized Access Only ^C
!
!
!
!
logging trap debugging
logging 10.0.1.71
line con 0
!
line aux 0
```



```
!  
line vty 0 4  
login  
!  
!  
ntp server 10.0.1.71  
!  
end
```

ISP : Building configuration...

Current configuration : 1000 bytes

```
!  
version 15.1  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
security passwords min-length 10  
!  
hostname ISP  
!  
!  
!  
!  
!  
!  
!  
!  
ip cef  
no ipv6 cef  
!  
!  
!  
!  
license udi pid CISCO2911/K9 sn FTX15249CM7-  
!  
!  
!  
!  
!  
!  
!  
!
```

```
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
ip address 209.165.200.225 255.255.255.240  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 64.100.1.1 255.255.255.224  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
ip address 64.100.2.1 255.255.255.224  
duplex auto  
speed auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
ip classless  
ip route 10.0.0.0 255.255.0.0 209.165.200.226  
ip route 192.168.2.0 255.255.255.0 64.100.1.2  
ip route 192.168.3.0 255.255.255.0 64.100.1.2  
ip route 192.168.10.0 255.255.255.128 64.100.2.2  
!  
ip flow-export version 9  
!  
!  
!  
banner motd ^C Authorized Access Only ^C  
!  
!  
!  
!  
line con 0
```

```
!  
line aux 0  
!  
line vty 0 4  
login  
!  
!  
!  
End
```

MIU_GW: Building configuration...

Current configuration : 3706 bytes

```
!  
version 15.1  
service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
security passwords min-length 10  
!  
hostname MIU-GW  
!  
!  
!  
!  
ip dhcp excluded-address 10.0.1.33 10.0.1.35  
ip dhcp excluded-address 10.0.1.49 10.0.1.51  
ip dhcp excluded-address 10.0.0.129 10.0.0.131  
ip dhcp excluded-address 10.0.0.65 10.0.0.67  
!  
ip dhcp pool VLAN10  
network 10.0.1.32 255.255.255.240  
default-router 10.0.1.33  
dns-server 10.0.1.70  
ip dhcp pool VLAN20  
network 10.0.1.48 255.255.255.240  
default-router 10.0.1.49  
dns-server 10.0.1.70  
ip dhcp pool VLAN130  
network 10.0.0.128 255.255.255.192  
default-router 10.0.0.129  
dns-server 10.0.1.70  
ip dhcp pool VLAN40  
network 10.0.0.64 255.255.255.192  
default-router 10.0.0.65  
dns-server 10.0.1.70
```

```
!  
!  
!  
no ip cef  
no ipv6 cef  
!  
!  
!  
!  
license udi pid CISCO2911/K9 sn FTX152424O7-  
license boot module c2900 technology-package securityk9  
!  
!  
!  
crypto isakmp policy 10  
encr aes 256  
authentication pre-share  
group 5  
!  
crypto isakmp key cisco123 address 64.100.1.2  
!  
!  
!  
crypto ipsec transform-set MIU_TS esp-aes 256 esp-sha-hmac  
!  
crypto map MIU_VPN_MAP 10 ipsec-isakmp  
set peer 64.100.1.2  
set transform-set MIU_TS  
match address VPN_ACL  
!  
!  
!  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet0/0  
ip address 209.165.200.226 255.255.255.240
```

```
ip nat outside
duplex auto
speed auto
crypto map MIU_VPN_MAP
!
interface GigabitEthernet0/1
ip address 10.0.1.94 255.255.255.252
ip nat inside
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface GigabitEthernet0/0/0
ip address 10.0.1.66 255.255.255.240
ip nat inside
!
interface GigabitEthernet0/1/0
ip address 10.0.1.82 255.255.255.252
ip helper-address 10.0.1.67
ip nat inside
!
interface GigabitEthernet0/2/0
ip address 10.0.1.90 255.255.255.252
ip helper-address 10.0.1.67
ip nat inside
!
interface GigabitEthernet0/3/0
ip address 10.0.1.86 255.255.255.252
ip helper-address 10.0.1.67
ip nat inside
!
interface Vlan1
no ip address
shutdown
!
router eigrp 10
redistribute ospf 100 metric 10000 100 255 1 1500
redistribute static
network 10.0.1.84 0.0.0.3
network 10.0.1.92 0.0.0.3
!
```

```
router ospf 100
router-id 1.1.1.1
log-adjacency-changes
redistribute eigrp 10 subnets
passive-interface GigabitEthernet0/0/0
network 209.165.200.224 0.0.0.15 area 0
network 10.0.1.80 0.0.0.3 area 0
network 10.0.1.84 0.0.0.3 area 0
network 10.0.1.88 0.0.0.3 area 0
network 10.0.1.92 0.0.0.3 area 0
network 10.0.1.64 0.0.0.15 area 0
default-information originate
!
ip nat pool MIU_PAT_POOL 209.165.200.227 209.165.200.227 netmask 255.255.255.240
ip nat inside source list NAT_ACL pool MIU_PAT_POOL overload
ip nat inside source static 10.0.1.69 209.165.200.228
ip nat inside source static 10.0.1.70 209.165.200.229
ip nat inside source static 10.0.1.68 209.165.200.230
ip classless
ip route 0.0.0.0 0.0.0.0 209.165.200.225
ip route 192.168.0.0 255.255.0.0 209.165.200.225
!
ip flow-export version 9
!
!
ip access-list extended VPN_ACL
permit ip 10.0.0.0 0.0.1.255 192.168.2.0 0.0.0.255
permit ip 10.0.0.0 0.0.1.255 192.168.3.0 0.0.0.255
ip access-list standard NAT_ACL
permit 10.0.0.0 0.0.1.255
permit 192.168.2.0 0.0.0.255
permit 192.168.3.0 0.0.0.255
ip access-list standard MIU_NAT
permit 10.0.0.0 0.0.255.255
permit 10.0.0.0 0.0.1.255
ip access-list standard NAT_ACL
permit 10.0.0.0 0.0.1.255
permit 192.168.2.0 0.0.0.255
permit 192.168.3.0 0.0.0.255
permit 192.168.10.0 0.0.0.127
!
banner motd ^C Authorized Access Only ^C
!
!
!
!
```

```
logging trap debugging
logging 10.0.1.71
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
ntp server 10.0.1.71
!
End
```

SW_S: Building configuration...

```
Current configuration : 1357 bytes
!
version 16.3.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
no profinet
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
```

```
!  
!  
!  
spanning-tree mode pvst  
!  
!  
!  
!  
!  
!  
interface GigabitEthernet1/0/1  
!  
interface GigabitEthernet1/0/2  
!  
interface GigabitEthernet1/0/3  
!  
interface GigabitEthernet1/0/4  
!  
interface GigabitEthernet1/0/5  
!  
interface GigabitEthernet1/0/6  
!  
interface GigabitEthernet1/0/7  
!  
interface GigabitEthernet1/0/8  
!  
interface GigabitEthernet1/0/9  
!  
interface GigabitEthernet1/0/10  
!  
interface GigabitEthernet1/0/11  
!  
interface GigabitEthernet1/0/12  
!  
interface GigabitEthernet1/0/13  
!  
interface GigabitEthernet1/0/14  
!  
interface GigabitEthernet1/0/15  
!  
interface GigabitEthernet1/0/16  
!  
interface GigabitEthernet1/0/17  
!  
interface GigabitEthernet1/0/18  
!
```



```
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
!
End
```

PINGS:

PC1 to PC7

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.1.2

Pinging 10.0.1.2 with 32 bytes of data:

Request timed out.
Reply from 10.0.1.2: bytes=32 time=12ms TTL=125
Reply from 10.0.1.2: bytes=32 time<1ms TTL=125
Reply from 10.0.1.2: bytes=32 time<1ms TTL=125

Ping statistics for 10.0.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 4ms

C:\>tracrt 10.0.1.2
Invalid Command.

C:\>tracert 10.0.1.2

Tracing route to 10.0.1.2 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      10.0.1.33
  2  0 ms      0 ms      0 ms      10.0.1.82
  3  0 ms      0 ms      0 ms      10.0.1.85
  4  0 ms      0 ms      0 ms      10.0.1.2

Trace complete.
```

Pc5 to pc12

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.6

Pinging 192.168.3.6 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.6: bytes=32 time<1ms TTL=124
Reply from 192.168.3.6: bytes=32 time<1ms TTL=124
Reply from 192.168.3.6: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.3.6:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 192.168.3.6

Tracing route to 192.168.3.6 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      10.0.0.129
  2  0 ms      0 ms      0 ms      10.0.1.90
  3  0 ms      0 ms      0 ms      209.165.200.225
  4  6 ms      0 ms      0 ms      64.100.1.2
  5  0 ms      0 ms      0 ms      192.168.3.6

```

Smart phone to Web server

\

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 209.165.200.228

Pinging 209.165.200.228 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 209.165.200.228: bytes=32 time=15ms TTL=125
Reply from 209.165.200.228: bytes=32 time=12ms TTL=125

Ping statistics for 209.165.200.228:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 15ms, Average = 13ms

C:\>tracert 209.165.200.228

Tracing route to 209.165.200.228 over a maximum of 30 hops:

  1  27 ms      9 ms      17 ms      192.168.10.1
  2  *          *          *          *
```

Pc9 to DNS server

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.1.70

Pinging 10.0.1.70 with 32 bytes of data:

Request timed out.
Reply from 10.0.1.70: bytes=32 time<1ms TTL=126
Reply from 10.0.1.70: bytes=32 time<1ms TTL=126
Reply from 10.0.1.70: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.1.70:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>tracert 10.0.1.70

Tracing route to 10.0.1.70 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.0.0.225
  2  0 ms    1 ms    0 ms    10.0.1.94
  3  0 ms    0 ms    0 ms    10.0.1.70

Trace complete.

C:\>
```